

- 1 (a) The Earth may be considered to be a uniform sphere of radius 6.38×10^6 m. Its mass is assumed to be concentrated at its centre.

Given that the gravitational field strength at the Earth's surface is 9.81 N kg^{-1} , show that the mass of the Earth is 5.99×10^{24} kg.

[2]

- (b) A satellite is placed in geostationary orbit around the Earth.

- (i) Calculate the angular speed of the satellite in its orbit.

angular speed = rad s^{-1} [3]

- (ii) Using the data in (a), determine the radius of the orbit.

radius = m [3]